

Local potential bases on Freudenthal meshes

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Abstract

Local bases for finite-element potential spaces provide the algebraic decompositions needed for vertex-patch corrections. On a consistently oriented Freudenthal mesh of the cube, we prove that continuous vector polynomials of degree $k + 1$ with continuous curl have a basis supported on single vertex stars for every $k \geq 5$ and every mesh size. This establishes the local-support assertion in the second conjecture of Farrell, Mitchell and Scott. The proof uses exact integer certificates together with analytical arguments covering all grids and degrees. At degree six, bounded windows transfer a fixed normal form and two-part local repairs. At higher degrees, clipped barycentric gaps reduce the construction to finitely many types, while affine inequalities verify their validity over unbounded parameter regions. We also obtain an exact dimension formula and provide reproducible basis-construction code. The conclusion is algebraic locality; a uniform bound on decomposition norms is a separate question.

1 Introduction

In a vertex-patch method, corrections act on the tetrahedra surrounding one mesh vertex. A potential supported on such a patch has a curl supported on the same patch, and that curl is divergence free. This gives a reason to ask whether the entire finite-element potential space can be generated by functions with this support. The question concerns the continuity of both a potential and its curl, so a local basis of the ordinary continuous polynomial space does not answer it.

Farrell, Mitchell and Scott formulate this question on uniform Freudenthal meshes in Conjecture 2(b) of [1, §3.3]. Their potential space consists of continuous vector-valued piecewise polynomials of degree $k + 1$ whose curl is continuous. They conjecture the existence of a basis supported on vertex stars when $k \geq 5$, and give numerical evidence. We use exactly this potential space, without imposed boundary conditions, on the six-tetrahedron subdivision specified in their Appendix A. Boundary vertices are included.

We prove the local-basis assertion for every integer $k \geq 5$ and every number $N \geq 1$ of cubes per coordinate direction. More precisely, each potential is a sum of potentials in the same space, each supported on one closed vertex star. Since the space is finite dimensional, a basis can be selected from these local generators. The construction acts on the potentials themselves; it does not identify potentials that differ by a gradient. Theorem 2.1 states the result and the dimension formula.

Two different finite constructions cover the polynomial degrees. For degree six, the coefficients and source equations of a fixed integer normal form fit within bounded windows. Boundary distances and coordinate residues determine which window to translate. Some canonical kernel vectors need to be split into two local potentials. For degrees at least seven, a fixed table describes all coefficient types. Large barycentric gaps remain unbounded variables,

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and elementary affine inequalities verify every source equation, pivot decision and support condition throughout its parameter region. Finite checks thus establish explicit families of identities rather than extrapolating ranks from selected meshes.

Inf-sup stability is a related but distinct issue. Ding, Ming, Yu and Zhu prove a divergence right-inverse estimate uniform in N for each fixed velocity degree $k \geq 4$, with pressure space equal to the actual divergence image [2, Theorem 2.1]. The construction here does not use that estimate. We prove algebraic locality without a uniform decomposition norm, and make no claim of multigrid convergence or exactness of a discrete Stokes complex.

Section 2 expresses the potential space as an integer matrix kernel and gives the support criterion. Section 3 proves the uniform constructions. Section 4 describes the implementation, certificate verification and exact construction cases.

2 Bernstein coefficients and local support

Let $\Omega = (0, 1)^3$, $N \geq 1$ and $h = 1/N$. For each $a \in h\{0, \dots, N-1\}^3$ and permutation $\sigma \in S_3$, put

$$T_{a,\sigma} = a + h \operatorname{conv}\{0, e_{\sigma_1}, e_{\sigma_1} + e_{\sigma_2}, (1, 1, 1)\}. \quad (1)$$

All cubes have the same orientation. The resulting mesh is $\mathcal{T}_h$. For a mesh vertex v , let ω_v be the union of its closed incident tetrahedra. For an integer $d \geq 6$ define

$$S_{N,d} = \{w \in C^0(\overline{\Omega}; \mathbb{R}^3) : w|_T \in \mathbb{P}_d(T)^3 \ (T \in \mathcal{T}_h), \quad \operatorname{curl} w \in C^0(\overline{\Omega}; \mathbb{R}^3)\}. \quad (2)$$

Here $\mathbb{P}_d$ denotes total degree at most d , and the curl is initially taken on each polynomial piece. No condition is imposed on a physical boundary face. Support is closed support in $\overline{\Omega}$.

Theorem 2.1. *For every integer $N \geq 1$ and $d \geq 6$, the real vector space $S_{N,d}$ has a basis each of whose members is supported in a single ω_v . Equivalently,*

$$S_{N,d} = \sum_v \{w \in S_{N,d} : \operatorname{supp} w \subseteq \omega_v\}. \quad (3)$$

For $N \geq 2$ its dimension is

$$3N^3(d-1)^2(d-2) + 3N^2(d-1)(5d-4) + 9N(2d-1) + 6. \quad (4)$$

At $d = 6$ this formula also holds for $N = 1$.

With $d = k + 1$, the dimension in (4) is $3N^3k^2(k-1) + 3N^2k(5k+1) + 9N(2k+1) + 6$. The support assertion includes all $N \geq 1, k \geq 5$. We prove the theorem in Section 3 by constructing matrix kernels with actual vertex-star support.

2.1 A global coefficient lattice

Use the scaled coordinates $y = Nx$, so mesh cells have side length one. The polynomial degree and support do not change, and $\operatorname{curl}_x = N \operatorname{curl}_y$. On an ordered tetrahedron with vertices $v_0, \dots, v_3$ in the scaled mesh, let λ_i be its barycentric coordinates. Its degree- d Bernstein polynomials are

$$B_\alpha^d = \frac{d!}{\alpha_0! \cdots \alpha_3!} \prod_{i=0}^3 \lambda_i^{\alpha_i}, \quad \alpha_i \in \mathbb{Z}_{\geq 0}, \quad |\alpha| = d.$$

They form a basis of $\mathbb{P}_d$: homogenizing any lower-degree monomial by $(\sum_i \lambda_i)^{d-r} = 1$ proves spanning, and their number $\binom{d+3}{3}$ equals the polynomial-space dimension. On a face opposite

v_j , exactly the terms $\alpha_j = 0$ remain and form its face Bernstein basis. Equality of traces is therefore equality of those coefficients.

Label a coefficient by

$$q = \sum_{i=0}^3 \alpha_i v_i \in \Lambda_{N,d} := \{0, \dots, Nd\}^3. \quad (5)$$

This is a coefficient label, not evaluation at q/d . To recover its canonical tetrahedron, set

$$a_j = \min(\lfloor q_j/d \rfloor, N-1), \quad r_j = q_j - da_j. \quad (6)$$

Sort r in decreasing order, breaking ties by increasing coordinate index, to obtain σ . The local multiindex is

$$\alpha = (d - r_{\sigma_1}, r_{\sigma_1} - r_{\sigma_2}, r_{\sigma_2} - r_{\sigma_3}, r_{\sigma_3}). \quad (7)$$

It is nonnegative and sums to d , proving that every label in $\Lambda_{N,d}$ occurs.

The vertices with positive α_i span the unique mesh entity $E(q)$ whose relative interior contains q/d . Every occurrence of q is the coefficient of that same entity, with its face-trace identification. Incident tetrahedra of an entity are connected through faces containing it, including at the boundary. Indeed, a sufficiently small ball about a relative-interior point meets only simplices containing that point. Its intersection with the cube is convex. Generic polygonal paths between incident tetrahedron interiors cross full faces and avoid the finite collection of lower-dimensional intersections. This gives the claimed connection. Consequently the trace identifications give precisely one scalar coefficient per q . Continuous vector polynomials are parametrized by $c = (c_{q,j})$, $j = 1, 2, 3$, with ambient dimension

$$n = 3(Nd + 1)^3. \quad (8)$$

2.2 Curl continuity and support

On a chain tetrahedron in scaled coordinates the barycentric gradients are

$$g_0 = -e_{\sigma_1}, \quad g_1 = e_{\sigma_1} - e_{\sigma_2}, \quad g_2 = e_{\sigma_2} - e_{\sigma_3}, \quad g_3 = e_{\sigma_3}. \quad (9)$$

Differentiation of the Bernstein expansion gives

$$d^{-1} \operatorname{curl}_y w = \sum_{|\beta|=d-1} B_\beta^{d-1} \sum_{i=0}^3 g_i \times c_{\sum_m \beta_m v_m + v_i}. \quad (10)$$

For an internal face and its degree- $(d-1)$ face multiindex β , subtract the two inner sums in (10). The three Cartesian components give integer rows of a matrix A_{all} ; its entries are $0, \pm 1, \pm 2$. Physical faces give no rows. Equal face traces also match curl values at edges and vertices by the preceding adjacency argument. Thus the coefficient representation identifies

$$S_{N,d} \simeq \ker_{\mathbb{R}} A_{\text{all}}. \quad (11)$$

Two components suffice on each internal face. For a continuous vector polynomial, the jump of each component's gradient is parallel to the face normal ν . Writing these multiples as a vector b , the curl jump is $\nu \times b$, perpendicular to ν . This polynomial identity holds for each face coefficient. If $\nu_t \neq 0$, the t th component is $-\sum_{j \neq t} (\nu_j / \nu_t)$ times the others. The normal may be chosen from (9), with nonzero components ± 1 . The matrix A obtained by omitting the first nonzero-normal component on each face therefore has the same real row space as A_{all} . Degree-six certificates use A ; higher-degree certificates verify all components.

For a label q , let $I(q)$ be the incident tetrahedra of $E(q)$ and put $J(q) = \bigcap_{T \in I(q)} \text{Vert}(T)$. A nonzero Bernstein coefficient makes the polynomial nonzero in every such tetrahedron, by local independence. Its support includes the closure of each of those interiors. It follows that

$$\text{supp } w \subseteq \omega_v \iff v \in \bigcap_{c_{q,j} \neq 0} J(q). \quad (12)$$

In particular, $v \in \text{Vert}(E(q))$ at each nonzero coefficient is sufficient. It is not asserted necessary: $J(q)$ may be larger on the boundary.

2.3 A normal form for the kernel

Lemma 2.2. *Suppose C has r rows and n columns, an ordered set of columns P satisfies $C[:, P] = I_r$, and $\text{row } C = \text{row } A$. If F is the complementary set of columns, then*

$$g_f = e_f - \sum_{p \in P} C_{p,f} e_p, \quad f \in F, \quad (13)$$

is a basis of $\ker_{\mathbb{R}} A$, of dimension $n - r$.

Proof. The equations $Cc = 0$ are $c_P = -C[:, F]c_F$. Free coordinates are arbitrary and determine c uniquely. The vectors (13) have the identity matrix in these coordinates, proving independence and spanning. $\square$

If every g_f is a sum of local vectors in the kernel, their combined parts form a finite local spanning set. Successively retain vectors independent of those already retained; the resulting subset is a local basis. For a small finite case we also use this elementary rank criterion: if integer matrices A, B satisfy $AB^T = 0$, A has a nonzero r -minor modulo a prime, B has a nonzero s -minor modulo that prime, and $r + s = n$, then B is a real kernel basis. The nonzero modular determinants are nonzero integers, giving real rank lower bounds; membership supplies the opposite dimension inequality. No equality of modular and real ranks is assumed.

3 Uniform constructions

3.1 Degree six: bounded windows and local repairs

For this subsection $d = 6$. The reference grid has $N = 6$. Reassembling the integer curl matrix gives a fixed certificate W, C, P satisfying

$$C = WA, \quad A = A[:, P]C, \quad C[:, P] = I. \quad (14)$$

Here $n = 151959$, $|P| = 72519$, and there are 79440 free columns. Define the window signature

$$\eta_R(q; N) = ((q_j \bmod 6, \min(q_j, R + 1), \min(6N - q_j, R + 1)))_{j=1}^3. \quad (15)$$

Distances and radii in this subsection refer to the coefficient lattice, so a scaled mesh vertex v is at coefficient position $6v$.

The following finite properties are verified for every row or column of the reference certificate. Section 4.2 specifies the checks and their arithmetic.

1. The pivot/free decision, and the relative C row at a pivot, depend only on $(\eta_5(q; 6), j)$. There are $17496 = 3 \cdot 18^3$ such decisions. Every nonzero coefficient of the row is within distance seven of its pivot in the infinity norm.

2. Every nonzero original row has its coefficients within distance two of its first coefficient position. The vertices of both incident tetrahedra lie within distance seven. For a source row used in $C_p = W_p A$, both full tetrahedra lie within distance eleven of the pivot.
3. Of the canonical vectors (13), 78715 have a recorded single-star owner. Each of the other 725 is the sum of two saved local kernel vectors. The 1450 repair parts have integer coefficients of magnitude at most two, and the sums have denominator one.
4. For every recorded owner v of a vector or a repair part anchored at free column f , $\|6v - q_f\|_\infty \leq 6$. Every vertex of its entire star lies within coefficient distance twelve of q_f .

The second property bounds the complete source geometry, which is stronger than bounding the positions of its nonzero coefficients.

Lemma 3.1. *For $N \geq 5$ and $R \in \{5, 7, 11, 12\}$, every signature $\eta_R(q; N)$ that occurs has a representative on $N = 6$. Their anchors differ by a vector in $(6\mathbb{Z})^3$. Translation identifies the mesh and its physical boundary within radius R . If two anchors have the same η_{R+s} , every shift of infinity norm at most s has the same domain-membership decision and, when inside, the same η_R at both anchors.*

Proof. In one coordinate the left and right bands of width R are disjoint, since $6N \geq 30 > 2R$. A point in the left band is represented by the same integer on $[0, 36]$; distance $t \leq R$ from the right boundary is represented by $36 - t$. For an interior point, the reference interval $[R + 1, 35 - R]$ contains all six residues. Residue equality makes the translation divisible by six. The exact distances identify each visible boundary plane; clipped distances put the others outside the window. For a shift of size at most s , a boundary at distance at most $R + s$ has its distance recorded exactly. A boundary farther away cannot enter the radius- R shifted window. Residues change by the same shift. Apply this argument to each coordinate. $\square$

Only occurring target signatures need representatives. At $N = 5, R = 12$ the interior interval contains five positions, so requiring all six residues there would be incorrect.

For any $N \geq 5$, define P_N and C_N by the reference η_5 table. To prove the forward inclusion, choose a reference pivot with the same η_{11} . Every source face and both adjacent tetrahedra in its identity $C_p = W_p A$ translate by Lemma 3.1. Their equations and weights are unchanged, proving $\text{row } C_N \subseteq \text{row } A_N$.

For the reverse inclusion, anchor an original nonzero row at its first coefficient and choose a reference with the same η_7 . This preserves the full row geometry. Each of its coefficient positions is within distance two, so its η_5 and pivot/free status are preserved. The relative C row at each pivot is therefore the same. The reference identity $a = a[:, P]C$ translates term by term, proving $\text{row } A_N \subseteq \text{row } C_N$. Finally choose η_{12} at a pivot: the radius-seven row and radius-five decisions at its coefficients agree with the reference. Thus its only pivot coefficient is one at the anchor, and $C_N[:, P_N] = I$. Lemma 2.2 gives a canonical kernel basis for every $N \geq 5$.

A canonical vector g_f depends on $\eta_{12}(q_f; N)$ and component. Indeed all its pivot coefficients are within distance seven of f ; their radius-five decisions and relative rows determine its entire pattern. Choose a reference free vector with that signature. If it is local, translate it and its owner; otherwise translate its two repair parts. Their complete owner stars fit within radius twelve and are matched by Lemma 3.1.

For a translated part, continuity inside the star follows from the reference face equations. At an internal face separating the star from its exterior, both the potential trace and curl trace of the reference part are zero, since its exterior polynomial is zero. Translation preserves the inside tetrahedron, that face, and whether it is physical. Extension by zero outside the target

star therefore preserves both continuities. The exterior tetrahedron need not be in the window: only the inside traces must be transported. Face connectivity gives continuity at edges and vertices as well. The repair sums remain exact equalities of potentials. These local vectors span every g_f , giving a local basis.

The small grids are handled by exact integer certificates and the minor criterion above. At $N = 1, 2, 3, 4$ the basis dimensions are respectively 795, 4164, 11913, 25842. Each basis has zero integer curl residual and nonempty support intersections. Modulo 1000003, the prescribed constraint minors are -1 , and the prescribed basis minors are $1, 1, -1, 1$, respectively. Their orders sum to the ambient dimension.

For $N \geq 5$, count the free η_5 types. Each boundary distance $0, \dots, 5$ occurs once at each end. In the interior interval $[6, 6N - 6]$, residue zero occurs $N - 1$ times and every other residue $N - 2$ times. Multiplying the coordinate multiplicities and summing the free types gives the exact polynomial

$$\dim S_{N,6} = 300N^3 + 390N^2 + 99N + 6. \quad (16)$$

The small-grid dimensions satisfy the same formula. This proves the degree-six part of Theorem 2.1.

3.2 Higher degrees: unbounded gap parameters

Here $N \geq 2, d \geq 7$. Use the canonical cell and gaps in (6)–(7). The cell boundary class b_j is zero for the first cell, two for the last, and one otherwise. Put $\gamma_i = \min(\alpha_i, 3)$ and define the coefficient type

$$\tau = (b, \sigma, \gamma, j), \quad j \in \{1, 2, 3\}. \quad (17)$$

For $\gamma_i < 3$ the gap is fixed to that integer; a value three means the unbounded variable $\alpha_i \geq 3$. Always $\sum_i \alpha_i = d \geq 7$.

The feasible types are determined exactly by three conditions. If there is no tail, $\sum_i \gamma_i \geq 7$. A zero intervening gap requires the tied consecutive ordering indices to increase. If $\gamma_0 = 0$, all leading coordinates whose remainder is d must be in a last cell. These conditions are necessary from the canonical rule. Conversely, choose the fixed gaps and tails, increasing a tail if needed to reach degree seven, and place the cell classes on $N = 3$. Reconstruct the remainders from the gaps. The tie conditions give the canonical order, and the last-cell condition makes every remainder equal to d valid. This realizes the type. Enumeration gives 59943 feasible types, and the rule table is checked to have exactly this set of keys.

The table declares 17331 pivot types and 42612 free types. Each pivot has a fixed integer relative row, with coefficient magnitude at most three and position offsets at most seven, and explicit source face equations. To verify these formulas over all degrees we use the following elementary minimum rule.

Lemma 3.2. *Let each integer x_i be fixed to ℓ_i or range over $x_i \geq \ell_i$, with $\sum_i x_i \geq s$. On a nonempty such region, an affine form $L = c_0 + \sum_i c_i x_i$ is unbounded below if a variable coefficient is negative. Otherwise its minimum is*

$$c_0 + \sum_i c_i \ell_i + \max(0, s - \sum_i \ell_i) \min_{i \text{ variable}} c_i. \quad (18)$$

If all coordinates are fixed, evaluate at that single feasible point. An affine identity holds throughout the region exactly when all variable coefficients vanish and the value at a feasible point is the desired constant.

Proof. Write $x_i = \ell_i + t_i$. Variable t_i are nonnegative and can increase independently. A negative coefficient makes L tend to $-\infty$. For nonnegative coefficients, place the required excess $\max(0, s - \sum \ell_i)$ entirely in a variable with least coefficient; this attains the lower bound. Independent increases also prove the identity criterion. $\square$

Forward row identities. Write $(R_\sigma \alpha)_{\sigma_i} = \sum_{t=i}^3 \alpha_t$ for $i = 1, 2, 3$, so $q = da + R_\sigma \alpha$. A source record lists two chain tetrahedra relative to a , their common face, an offset δ , a curl component and an integer weight. Each tetrahedron must have the three distinct positive coordinate steps and the listed face must be exactly their three-vertex intersection. Its relative vertices obey coordinate bounds $[0, 2]$ in a first cell, $[-1, 2]$ in an interior cell, and $[-1, 1]$ in a last cell. These bounds keep the source in the physical mesh for all $N \geq 2$ in that context.

Write its relative barycentric functions as $\lambda_i(z) = \ell_i + g_i \cdot z$. The source's face multiindex at $(d-1)a + R_\sigma \alpha + \delta$ is

$$\beta_i = \ell_i(d-1) + g_i \cdot (R_\sigma \alpha + \delta). \quad (19)$$

Lemma 3.2 checks that the opposite coordinate is identically zero and the three face coordinates are nonnegative throughout the type's unbounded region. They sum to $d-1$ since $\sum \ell_i = 1$ and $\sum g_i = 0$. Each source is therefore an actual face equation for every admissible degree. Substituting (10) and the integer weights must give the declared row exactly, after summing duplicate offsets. All 105320 sources satisfy these checks, including 631920 nonnegativity inequalities and 210640 plane identities. This proves $\text{row } C \subseteq \text{row } A_{\text{all}}$ for the matrix obtained by applying the table at each coefficient site.

Shifted types and pivots. For a row of offset radius $\rho \leq 7$, set $M = \max(7, 2\rho + 3) \leq 17$. Partition every tail into the alternatives $\alpha_i = 3, \dots, M-1$ and $\alpha_i \geq M$, retaining $\sum \alpha_i \geq 7$ and discarding empty regions. For a shifted site $q' = q + \delta$, classification at a feasible representative proposes its cell a' , order and clipped gaps. The proposal is then proved throughout the region by Lemma 3.2, checking

$$\begin{aligned} 0 \leq q'_j \leq Nd, \quad r'_j = q'_j - da'_j \geq 0, \\ d - r'_j \geq 1 \quad (a'_j < N-1), \quad d - r'_j \geq 0 \quad (a'_j = N-1). \end{aligned}$$

Consecutive remainders must be nonincreasing, with a difference at least one when their indices are decreasing. A proposed gap below three must be identically that integer; a tail must be at least three everywhere. Thus a representative is never used as a substitute for a uniform check.

Since $\rho \leq 7 \leq d$, a shifted canonical cell differs by at most one per coordinate. For $N \geq 4$, first, second, distant interior, penultimate and last cells represent all possible boundary contexts. They are realized on $N = 5$ at positions 0, 1, 2, 3, 4; $N = 4$ simply lacks the distant case. The boundary status of every one-cell shift is preserved. The overlapping contexts for $N = 2, 3$ are processed directly. Cartesian products of these contexts cover all grids.

Every checked off-diagonal entry is at a free type, and the only pivot entry of each row is one at its anchor. The finite traversal covers 249177 parameter regions, 1569530 grid-context regions and 10782184 off-diagonal links. This proves $C[:, P] = I$ uniformly.

Vertex owners. The positive-gap vertices of the canonical tetrahedron are $E(q)$; their identity depends only on σ, γ . Start the owner mask of a free type with these vertices, expressed as the eight binary vertices of its canonical cube. For every link $C_{p,f} \neq 0$, intersect that mask with $\text{Vert}(E(q_p))$ expressed relative to the free site's cell. Intersect over all regions and all boundary contexts just described. Every possible link is covered, and all 42612 free-type masks remain nonempty. A surviving vertex belongs to $E(q_f)$ and every $E(q_p)$ occurring in g_f . Choosing its least bit gives a mesh vertex v_f . The sufficient condition in (12) proves $\text{supp } g_f \subseteq \omega_{v_f}$ for every grid and degree.

Reverse row identities. For an internal face and its two tetrahedra let o be their coordinatewise minimum and let w_j be their coordinate widths. The tets occupy one cube or two

cubes sharing a square face, so $w_j \in \{1, 2\}$. Record the relative face and tetrahedron vertices and the boundary information

$$(\mathbf{1}_{o_j=0}, \min(N - o_j - w_j, 2))_{j=1}^3. \quad (20)$$

Enumeration on $N = 2, \dots, 6$ gives exactly 888 distinct profiles, and a set comparison verifies the file against that enumeration. This covers every N : for $N \geq 6$ map each origin coordinate to $N = 6$ by

$$o'_j = \begin{cases} 0, & o_j = 0, \\ 6 - w_j, & N - o_j - w_j = 0, \\ 5 - w_j, & N - o_j - w_j = 1, \\ 1, & \text{otherwise.} \end{cases}$$

When $o_j = 0$ its upper distance is at least four, so the first case preserves both clipped distances. All other cases preserve the lower flag and the upper distance class. Smaller grids were enumerated directly.

Let the face multiindex be $\beta = (\beta_0, \beta_1, \beta_2)$, with $d = 1 + \sum \beta_i \geq 7$. If f_i and t_j are relative face and tetrahedron vertices, an original row coefficient occurs at

$$q = do + \sum_i \beta_i f_i + t_j. \quad (21)$$

Its position q/d is a convex combination of actual vertices. Its canonical cell lies within the two tetrahedra's bounding box; profile (20) consequently preserves whether that cell is first, interior or last, including upper-plane coefficients assigned to the last cell. Relative remainders and ordering are preserved by cell translation.

Partition each β_i into 0, 1, 2, 3, 4 and the unbounded tail $\beta_i \geq 5$, retaining $\sum \beta_i \geq 6$. On every feasible region, classify all nonzero columns of every Cartesian curl row using the uniform affine tests. Subtract the appropriate C row at each pivot column, with multiplier equal to that original coefficient. Store the residual by relative coefficient offset and component, subtracting the common moving face position. Each residual is the empty integer dictionary. Hence the exact identity

$$a = a[:, P]C \quad (22)$$

holds throughout each region. The checks cover 144744 regions, 358926 nonzero row identities and 1713456 coefficient classifications. Zero original rows are automatic identities. All three curl components are included. We obtain $\text{row } A_{\text{all}} \subseteq \text{row } C$ for all $N \geq 2, d \geq 7$.

The forward and reverse identities and unit pivot block now give a kernel basis by Lemma 2.2. The owner intersections prove that each member is supported on one vertex star. This proves the support part of Theorem 2.1 in the higher-degree range.

3.3 Counting and completion of the theorem

For a free type, let b be the number of interior cell-coordinate classes, m the number of tail gaps, and $s = \sum_i \gamma_i$. Its cell multiplicity is $(N - 2)^b$. For $m > 0$ the number of gap choices at degree d is

$$\binom{d - s + m - 1}{m - 1} \quad (d \geq s),$$

and zero otherwise: write each tail as $3 + t_i$ and distribute $d - s$ nonnegative units. For $m = 0$ the multiplicity is one if $d = s$ and zero otherwise. Canonical classification makes this a bijective count.

All lower sums are at most twelve, and types without tails have sum at most eight. For $d \geq 12$ the dimension is therefore the polynomial

$$\sum_{\tau \text{ free, } m > 0} (N-2)^b \frac{\prod_{t=1}^{m-1} (d-s+t)}{(m-1)!}. \quad (23)$$

Exact rational expansion of the fixed table gives (4). For $d = 7, \dots, 11$, the binomial/indicator sum is checked as an exact polynomial in N to give the same formula; $d = 12$ is checked at the transition as well. Equation (16) supplies degree six.

Completion of Theorem 2.1. The degree-six window construction, including the four small-grid certificates, proves (3) for all N at $d = 6$. The gap construction proves it for $N \geq 2, d \geq 7$. If $N = 1$, every tetrahedron contains $(0,0,0)$, whose star is the whole cube. The space is finite dimensional, so any basis proves the remaining cases. Selecting an independent subset of local generators preserves support. The counts above prove the stated dimension ranges. Finally $y = Nx$ preserves support and both continuities, completing the physical-domain statement. $\square$

4 Implementation and exact experiments

4.1 Basis construction

The accompanying Python package stores each basis member as a row of integer vector Bernstein coefficients. Its input is $N \geq 1, d \geq 6$; the output is a sparse basis matrix B , one owner vertex per row, and, when available, the normal form and pivot/free columns. Certificate files are verified against their SHA-256 manifest before use. Components are indexed from zero in the code, with increasing-index tie breaks as above.

Algorithm 1. Local potential basis construction

Input: integers $N \geq 1, d \geq 6$; fixed certificate tables.

Output: $B \in \mathbb{Z}^{D \times n}$ and D owner vertices.

Build the mesh, lattice labels and local Bernstein maps; $n = 3(Nd + 1)^3$.

if $d = 6$ and $N < 5$:

 Load the finite local basis and compute its support intersections.

else if $d = 6$:

 Apply the η_5 table to all n coefficients; form C, P, F .

 Form each g_f by (13); choose its $N = 6$ representative by η_{12} .

 Keep a local g_f , or replace it by its two translated repair parts.

 Select independent rows using exact elimination modulo 1000003.

else if $N \geq 2$:

for each coefficient (q, j) , classify τ and append its fixed row or free index.

 Form B from (13); assign each free type's least owner-mask bit.

else:

 Compute an integer kernel on the one-cube grid; assign owner $(0, 0, 0)$.

 Reassemble all three curl components and require $A_{\text{all}} B^T = 0$.

 Check every recorded owner, independence and the applicable dimension count.

 Return the checked sparse rows and owner vertices.

For $N \geq 2, d \geq 7$, construction requires $O(n)$ coefficient operations and storage in a unit-cost model with constant-time lattice addressing. The type table and maximum offsets

are fixed. Each canonical vector contains only sites within radius seven, so its number of coefficients is bounded independently of N, d . The explicit tetrahedral Bernstein map has $6N^3 \binom{d+3}{3} = O(n)$ entries. The implementation uses dictionary addressing, for which this cost is expected rather than a worst-case hash bound. These estimates exclude full certificate replay and bit complexity of arbitrarily large indices.

At degree six, translating the local generating family also has $O(n)$ size. The independent-subset selection may create fill during elimination; we do not assign it the same linear cost. A dense worst-case bound is $O(D^3)$ arithmetic and $O(D^2)$ workspace for the projected generator matrix, where $D = \dim S_{N,6}$ and there are at most $2D$ generators. Because each canonical vector is an integral sum of those generators, their free-coordinate projections span modulo any prime. Nonzero modular minors imply real independence, while exact integer membership is checked separately. The one-cube implementation uses modular elimination with a rational lift and accepts it only after exact membership and complementary rank bounds; unsuccessful reconstruction raises an error.

The package checks integer accumulation bounds before sparse products and raises an error rather than allowing possible overflow. Pointwise potential and physical-curl evaluation uses floating-point Bernstein formulas and is tested separately against a global polynomial field. No conditioning or numerical-stability theorem is asserted for evaluation.

4.2 Finite witness verification

The proof's fixed witnesses accompany the submission package *AIM 633: Local potential bases on Freudenthal meshes*, version 0.1.0. The directory `src/freudenthal/certificates/` contains `all_degree_rules.json.gz` and `verification_data.zip`, with respective SHA-256 hashes:

```
43d45cfd9394befe130b7e5e27f22851369e0adaf0669f8526e3d8156bc7b5ee
32c3c6847241288a4607d699e68ac7fbbdae3378e4057531c44a94274f85f7c7
```

The release manifest verifies these files and the ZIP's member manifest identifies every extracted witness. From the installed repository, run `python verify_all.py` to replay the checks in a fresh local result directory. These are essential proof data, distinct from the larger generated bases. The verification scripts check the following statements directly:

1. Degree six: reconstruct the matrices; verify all identities in (14); traverse every relative normal-form signature; bound complete source geometries; reconstruct the canonical vectors; check every repair's membership, owner star and exact sum; verify the four prescribed small-grid minors.
2. Higher degrees: compare rule keys with the complete feasible-type set; prove source admissibility using (19); verify exact integer source sums; traverse all gap and boundary contexts for the unit pivots and owner intersections; compare the face-profile set with its geometry enumeration and verify (22) on every unbounded face-parameter region.
3. Count the free types using exact rational polynomial arithmetic, including the transition degrees. Independently reassemble degree-six matrices with the analytical gradients (9).

The normal-form discovery code and recovered rank observations are not needed to accept any of these predicates.

Python geometry, source and affine checks use exact integers; dimension expansion uses exact rationals. Sparse products are evaluated as bounded integer sums. The C++ affine checker uses signed 64-bit integers and is compiled with undefined-behavior sanitization. It evaluates bounded affine representatives, not an unbounded actual degree. In the pivot checks each representative gap is at most seventeen and the degree at most sixty-eight; reference grids

have at most six cells per axis. Classification-form coefficients are bounded by twenty and their evaluations by two thousand. Even the permitted input bounds of 1000 row terms and coefficient magnitude 1024 give a conservative residual accumulation bound $16 \cdot 2 \cdot 1000 \cdot 1024 < 2^{63} - 1$. The fixed run's loop counters are below 10^9 . Python integer arithmetic checks the JSON-to-integer table export before the C++ traversal.

These checks form a computer-assisted proof using an inspected program, integer arithmetic libraries, compiler and hardware as its trusted computing base. The analytical window and parameter-coverage arguments establish why the finite traversals cover all N, d . The work is not a proof-assistant formalization.

4.3 Construction cases and a sparsity diagnostic

Table 1 reports ten constructions. All coefficients are integers; membership residuals are exactly zero, without a floating tolerance. Each case checks all three curl components, every owner-star condition and independence. Cases at the small-grid boundary, on both sides of the degree-six reference grid, and in the higher-degree range exercise different branches. They are implementation regressions, in addition to the uniform certificates, rather than a substitute for the parameter-coverage proof.

Table 1: Exact construction cases. $d = k+1$ is the potential degree; n is the ambient coefficient count, D the basis dimension and $\text{nnz}(B)$ the number of nonzero scalar component entries in the Bernstein representation. Every all-component curl residual has zero nonzero entries and every reported basis row has a verified single-star owner.

N	d	n	D	$\text{nnz}(B)$
1	6	1029	795	3072
2	6	6591	4164	27565
3	6	20577	11913	97975
4	6	46875	25842	238875
5	6	89373	47751	478780
7	6	238521	122709	1342470
2	7	10125	6792	35712
4	7	73167	43962	307104
3	13	192000	148749	421569
2	20	206763	178548	335622

An independent modular rank calculation on $N = 2, d = 7$ at both primes 1000003 and 1000033 agrees with the constructed dimension. Evaluation is checked against $w = (z^2, x^2, y^2)$, represented in degrees six, seven and twenty, with physical curl $(2y, 2z, 2x)$. Degree-six traces are also compared from both sides of every internal face on $N = 2$. Evaluation tests use double precision, absolute tolerance $2 \cdot 10^{-14}$ and relative tolerance $2 \cdot 10^{-13}$ for the polynomial values and curl; the two-sided comparison uses 10^{-13} and 10^{-12} , respectively.

To describe the sparse representation, define

$$F(s) = D^{-1} \# \{i : \text{nnz}(B_{i,:}) \leq s\}.$$

The count is over scalar columns (q, j) , rather than grouping the three components at one label q . Figure 1 uses the complete integer histograms for $N = 4$ at degrees six and seven. A larger $F(s)$ means more basis members fit within that coefficient budget. The mean counts are approximately 9.24 and 6.99, respectively, and both maxima are 69. This is a description of the constructed bases, not a bound on their condition numbers or on decomposition norms.

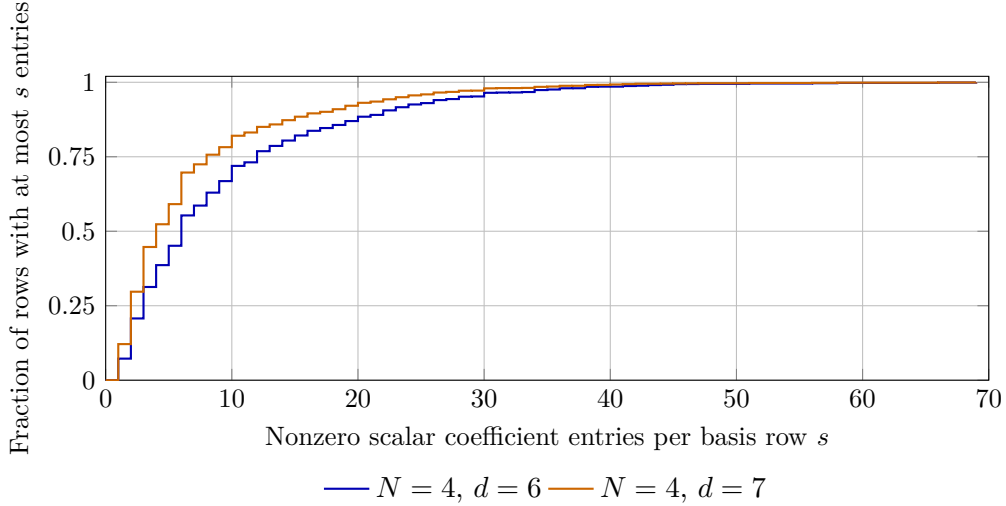


Figure 1: Cumulative coefficient counts for the local bases on $N = 4$. The full histograms are normalized by their own basis dimensions, with the same scalar-entry-budget axis and no subsampling or clipping of counts. The curves describe sparsity of the chosen Bernstein representations; they do not measure accuracy or solver convergence.

The accompanying `src` package, `test` suite and `numerical_case` scripts reproduce the certificate checks, Table 1 and Figure 1. Compact count summaries and vector figures are retained with the code. Full generated matrices, command logs, input hashes and environment manifests are kept in local result directories and regenerated by the documented commands. Thus the distributed finite witnesses suffice to verify the theorem without retaining every generated numerical basis.

Algebraic locality leaves a quantitative question: can a decomposition in (3) be chosen with a bound uniform in N in a suitable potential norm? Coefficient sparsity alone does not provide such a bound. That estimate and its use in multigrid require further analysis.

References

- [1] P. E. Farrell, L. Mitchell and L. R. Scott, *Two conjectures on the Stokes complex in three dimensions on Freudenthal meshes*, SIAM Journal on Scientific Computing **46**(2) (2024), A629–A644. doi:10.1137/22M1533943. Primary text: [arXiv:2211.05494v2](https://arxiv.org/abs/2211.05494v2).
- [2] S. Ding, P. Ming, H. Yu and Q. Zhu, *The Scott–Vogelius element is inf-sup stable on Freudenthal meshes for $k \geq 4$* , preprint, 22 September 2026. [arXiv:2609.25690v1](https://arxiv.org/abs/2609.25690v1).

AI-use declaration

ChatGPT assisted with the candidate construction and certificate-producing code. Codex assisted with the expanded proof, reference verification, implementation, tests and manuscript preparation. All reported finite certificate checks and construction cases were replayed from the saved data; no proof-assistant verification or independent human review is claimed.